

Military Cyber OSINT Collection Tool

How it works — explained from scratch, in plain language

1. What is this project, in one sentence?

It is a program that automatically checks dozens of public, legal information sources every day to see whether any India-related (and neighbouring-country) government or military infrastructure is accidentally exposed on the internet in a way that could be a security risk — then organises everything it finds into one clear, prioritised report.

What does “OSINT” mean?

OSINT stands for **Open-Source Intelligence**. It simply means gathering information from sources that are already public and legal to look at — things like search engines built for scanning the internet, public certificate records, public breach databases, and public news feeds. Nothing in this tool involves hacking, breaking into a system, or accessing anything private. It only looks at information that is already visible to anyone who knows where to look — the tool's job is to look in all those places at once, automatically, instead of a person doing it manually.

Think of it like this: It's the digital version of a security guard walking around a building at night checking which doors and windows were accidentally left unlocked — not breaking in, just noticing and reporting what's already visible from the outside.

Why does this matter?

Government and defence organisations run huge numbers of websites, servers, and email systems. It is very easy for one of these to be set up correctly at first and then quietly become exposed later — for example, a test server left open, an old certificate exposing a hidden subdomain, or an admin login page left reachable from the public internet. Attackers already use these same public sources to find such weaknesses. This tool exists so that the exposure is found and reported to the defensive side first.

2. The big picture: how data flows through the tool

Every time the tool runs, it moves through four stages, always in the same order:

Stage	What happens
1. Collect	About 30 different public sources are each queried automatically, looking for anything related to a fixed list of known Indian and neighbouring-country government/military domains.
2. Clean up	Results that are irrelevant, duplicated, or already reported in the last 30 days are filtered out, so only genuinely new and relevant findings remain.
3. Cross-check	Findings are compared against each other. If several independent sources point to the same thing, that finding is trusted more and marked more serious.

Stage	What happens
4. Report	Everything that survives is written out as a searchable dashboard, a spreadsheet, and two extra report formats for sharing (explained in Section 5).

Think of it like this: Imagine hiring about 30 different specialists — one who only checks certificates, one who only checks leaked passwords, one who only checks exposed cameras and servers, and so on. Each specialist reports back separately. Instead of handing you 30 separate, messy reports, someone combines them, throws out anything not actually relevant, and flags anything that several specialists agree on. That combining person is what stages 2 and 3 do automatically.

3. Where the data actually comes from

The sources fall into a few natural groups. Not every source needs a paid subscription — most are free, and the tool simply skips a source if it isn't configured, so nothing breaks.

Category	Examples used	What it tells us
Internet device search engines	Shodan, Censys, ZoomEye, Netlas.io	These constantly scan the entire internet and remember what they find — open ports, running software, banners. Searching them tells us what a target's servers look like from the outside, without us scanning anything ourselves.
Certificate records	crt.sh	Every time a website sets up secure HTTPS (the padlock icon), that event is recorded permanently in a public, tamper-proof log. Searching these logs reveals subdomains an organisation may not realise are publicly listed.
Leak / breach checkers	Have I Been Pwned, LeakIX, GrayhatWarfare, Hudson Rock	These check whether staff email addresses, passwords, or files from an organisation have already turned up in a known data breach or an accidentally-exposed cloud storage bucket.
Threat intelligence feeds	AlienVault OTX, ThreatFox, MalwareBazaar, CISA, NVD	Public databases maintained by the security community listing known malware, attacker infrastructure, and dangerous software vulnerabilities (CVEs).
News & chatter monitoring	Defence news RSS feeds, public Telegram channels	Keeps track of relevant news coverage and public discussion that might relate to a tracked organisation.
Dark web monitoring	Tor-based search, ransomware leak-site trackers	Checks hidden parts of the internet (only reachable through the Tor network) for mentions of tracked organisations — e.g. a ransomware group listing a victim.

4. The “smart” parts — how it decides what matters

Collecting raw data from 30 sources would just create noise. Most of the actual engineering in this tool is in narrowing that raw data down to what's genuinely worth a person's attention. It does this with a few

clear rules rather than a black-box AI model.

a) Deciding if something is actually relevant

Every result is checked against a tiered set of rules before it's kept: does it belong to one of the tracked government/military domains directly? If not, does its content contain strong military-specific terms or known defence-contractor/attacker-group names? If not, does it contain enough weaker supporting terms to still be worth flagging? Anything that fails all three checks is discarded rather than cluttering the report.

b) Trusting findings that agree with each other

If the exact same server or address is independently flagged by three or more separate sources, the tool automatically raises that finding to the highest severity level ("CRITICAL"). One source saying something might be a coincidence; three unrelated sources agreeing almost never is.

c) Not repeating itself

The tool remembers every finding it has already reported in the last 30 days. On the next run, anything already reported recently is silently skipped, so each day's report only shows what's genuinely new — not the same 900 rows repeated every single day.

d) Recognising what software is actually running

When a server reveals information about itself (a "banner"), the tool checks it against a list of well-known software products (mail servers, admin panels, VPN gateways, etc.) and tries to pull out the exact version number too. Old, outdated software versions are a well-known security risk, so recognising "this is Product X, version from 2019" is far more useful than just "a server responded."

e) Scoring how serious a finding is

Severity (Critical / High / Medium / Low) isn't guessed — it's built up from several independent signals: how well-known and dangerous the software is, whether other security services report the same IP address as malicious, how exploitable a related vulnerability is (using an industry statistic called EPSS), and how many sources agree. A response that reveals nothing useful (for example, a server just saying "access denied" with no further detail) is deliberately scored lower than one that reveals a real, identifiable, outdated product.

***Think of it like this:** This is similar to how a doctor doesn't diagnose from one symptom alone — they combine several signs (temperature, blood pressure, test results) before deciding how serious a case is. The tool does the same thing with several independent security signals instead of trusting any single one blindly.*

5. A real example of the tool being improved

To make this concrete: one of the internet-scanning sources (Shodan) was recently reviewed by hand, row by row, to check whether its results were actually trustworthy. Two clear problems were found and fixed:

- **Generic, content-free results were being over-rated.** Several results were nothing more than a cloud provider's load balancer saying "access denied" — no real information at all — yet every single one was being marked as high severity. These are now automatically scored lower.

- **Some results didn't actually belong to the organisation being searched for.** A search for one country's Ministry of Defence returned a server that, when checked, turned out to belong to a completely unrelated commercial mail service — a case of the search engine's own records being out of date. The tool now cross-checks each result's own self-reported identity against the domain that was actually searched for, and flags any mismatch instead of trusting it blindly.

A second improvement added a brand-new, free, and completely independent check: for every IP address found by any source, the tool now also looks up *who actually owns that address space* (via a public internet-registry lookup called RIPEstat). This caught, for example, several “exposed border-security infrastructure” results that turned out to simply be generic Amazon cloud address space, not anything actually owned by that organisation — independent confirmation of the very same problem found by hand in the first fix.

This is a useful example to describe in a review: the tool isn't just collecting data, it is being actively audited and corrected against real evidence.

6. What comes out at the end

Output	What it's for
Dashboard (a web page)	The main way to actually browse results — searchable and filterable by category, severity, source, and more, with summary charts at the top.
Master spreadsheet (CSV)	One running database of every finding ever recorded, used by the dashboard and usable in Excel directly.
STIX file	An industry-standard file format for sharing threat intelligence. Any professional security tool that understands this format (used across the cybersecurity industry) can import this file directly — it isn't locked into this tool alone.
Weekly delta report	A short plain-text summary: how many genuinely new findings this week compared to last week, broken down by category and severity — useful for a quick status update.

7. Why this is useful

Manually checking even a handful of these ~30 sources for a single domain, every day, would take a person a significant amount of time — and this tool tracks a whole list of domains across India and several neighbouring countries at once. What it produces isn't raw, unfiltered data; it is the end result of relevance-checking, cross-source verification, de-duplication, software-version recognition, and multi-signal severity scoring — the same kind of judgement calls a human analyst would make, applied consistently and automatically, every single run.

In short: it turns scattered, publicly-available information into one organised, prioritised, and continuously updated security picture.